The Economics of Density

Evidence from the Berlin Wall

Ahlfeldt, Redding, Sturm and Wolf (2015)

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Motivation and Background

Models of Cities

- Your friend and mine, the Alonso–Muth–Mills model: monocentric city, bid-ask gradients, lots of fun, way too abstracted to make precise predictions.
- Some extensions: assorted Fujita & Krugman throughout the 90s, Lucas & Rossi-Hansberg (2003) give more flexibility in firm location and allow for agglomeration and location-varying productivity but only on a simple disk

Issues with These Models

These models do not capture important features of cities:

- Real-world, non-radial transport networks
- Local amenities and production externalities for residents
- Dispersed employment across cities

Contributions of This Paper

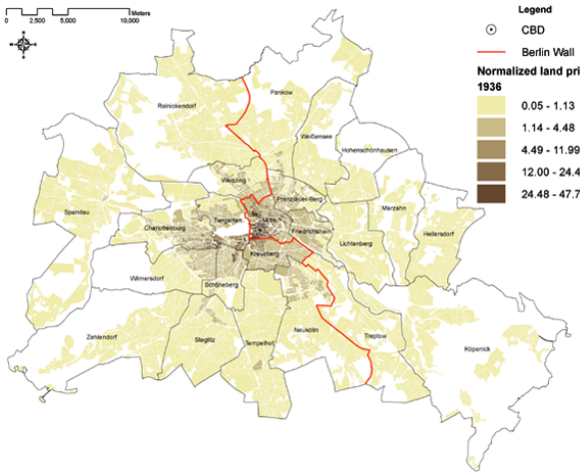
- A tractable quantitative spatial model (QSM) for counterfactuals and policy analysis.
- Estimates of agglomeration externalities and other highly important parameters
- CRAZY data work

The Berlin Wall: Timeline

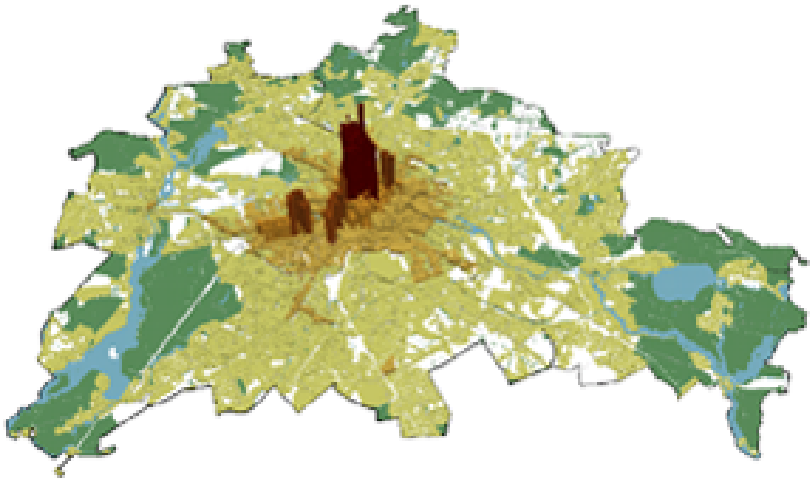
- **1945–1949:** Berlin divided, separate East and West German states established in 1949.
- **1961:** East Germany closes the border and builds the wall, ending commuting between East and West Berlin.
- **1989:** The wall comes down.

Data

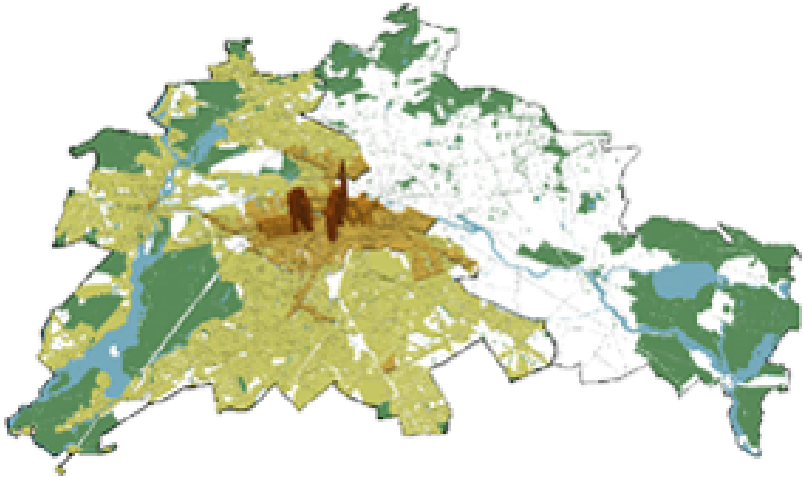
Land Prices in Berlin, 1936



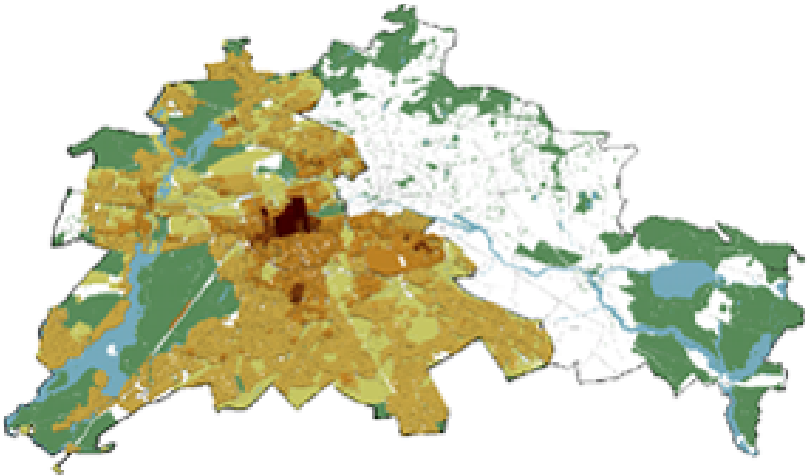
Land Prices: Berlin, 1936



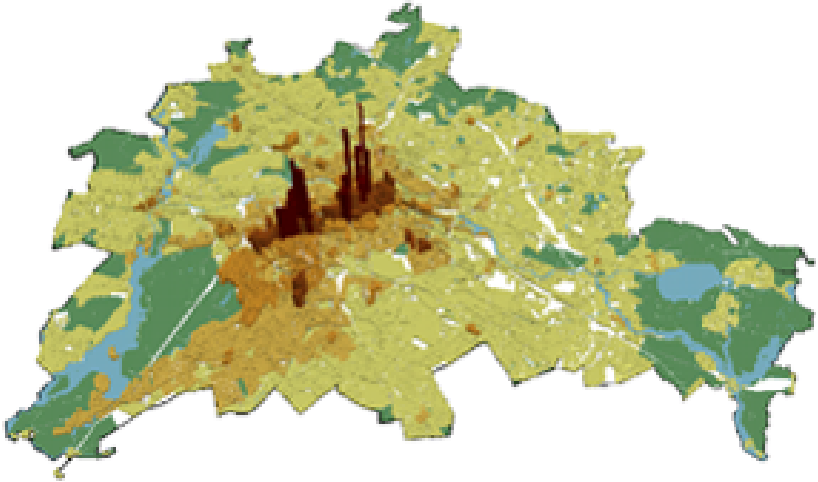
Land Prices: West Berlin, 1936



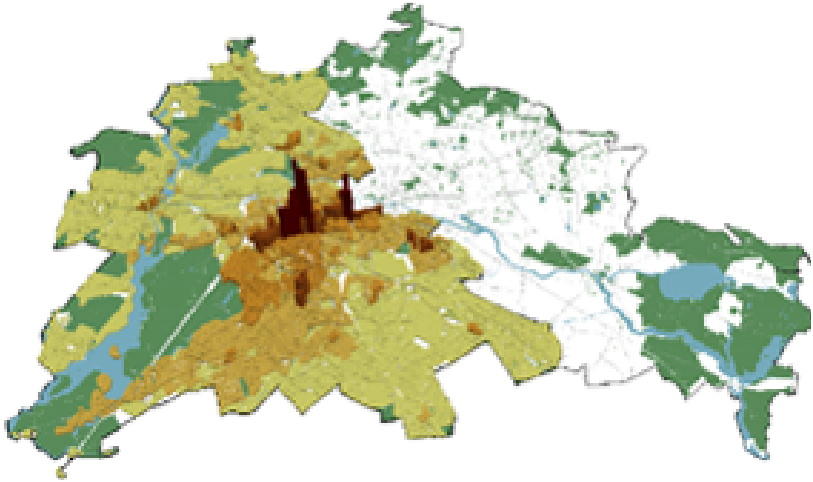
Land Prices: West Berlin, 1986



Land Prices: Berlin, 2006



Land Prices: West Berlin, 2006

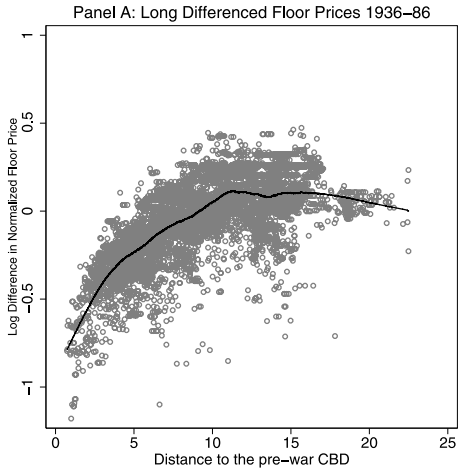


Other Key Data Sources

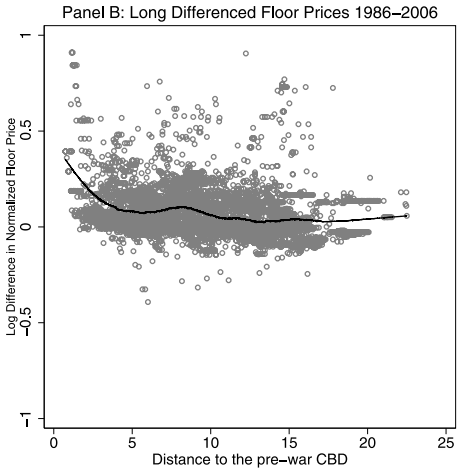
- **Workplace employment:** 1933 and 1987 censuses, the 1931 Berlin company register, and 2003 social-security employment records.
- **Resident employment:** census data and 2005 block population records. 1933 aggregate employment disaggregated spatially using a geocoded complete firm register (!?) and locations of public sector buildings
- **Travel times:** historical rail, tram, bus and road networks, used to calculate journeys between blocks.
- **Commuting patterns:** surveys from 1936, 1982 and 2008. No cars in 1936

Reduced Form Estimates

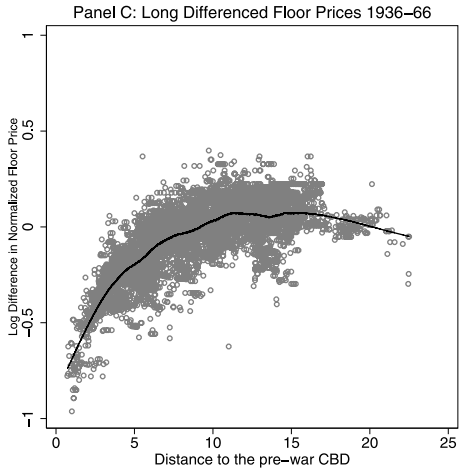
Division, 1936–1986



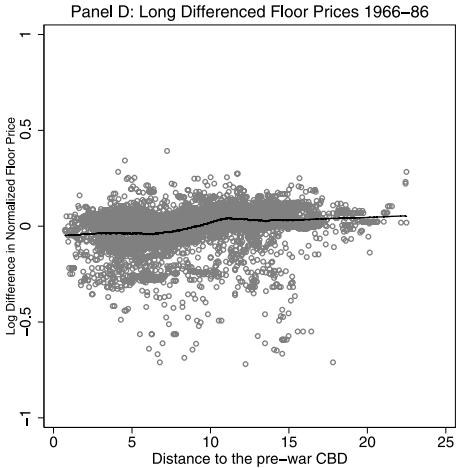
Reunification, 1986–2006



Division, 1936–1966



Placebo, 1966–1986



Placebo, 1928–1936

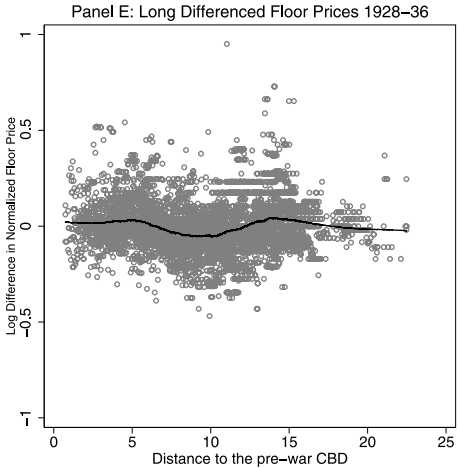


Table 1: Division, 1936–1986

	(1) $\Delta \ln Q$	(2) $\Delta \ln Q$	(3) $\Delta \ln Q$	(4) $\Delta \ln Q$	(5) $\Delta \ln Q$	(6) $\Delta \ln EmpR$	(7) $\Delta \ln EmpR$	(8) $\Delta \ln EmpW$	(9) $\Delta \ln EmpW$
CBD 1	-0.800^{***}	-0.567^{***}	-0.524^{***}	-0.503^{***}	-0.565^{***}	-1.332^{***}	-0.975^{***}	-0.691^{*}	-0.639^{*}
	(0.071)	(0.071)	(0.071)	(0.071)	(0.077)	(0.383)	(0.311)	(0.408)	(0.338)
CBD 2	-0.655^{***}	-0.422^{***}	-0.392^{***}	-0.360^{***}	-0.400^{***}	-0.715^{**}	-0.361	-1.253^{***}	-1.367^{***}
	(0.042)	(0.047)	(0.046)	(0.043)	(0.050)	(0.299)	(0.280)	(0.293)	(0.243)
CBD 3	-0.543^{***}	-0.306^{***}	-0.294^{***}	-0.258^{***}	-0.247^{***}	-0.911^{***}	-0.460^{**}	-0.341	-0.471^{**}
	(0.034)	(0.039)	(0.037)	(0.032)	(0.034)	(0.239)	(0.206)	(0.241)	(0.190)
CBD 4	-0.436^{***}	-0.207^{***}	-0.193^{***}	-0.166^{***}	-0.176^{***}	-0.356^{**}	-0.259	-0.512^{***}	-0.521^{***}
	(0.022)	(0.033)	(0.033)	(0.030)	(0.026)	(0.145)	(0.159)	(0.199)	(0.169)
CBD 5	-0.353^{***}	-0.139^{***}	-0.123^{***}	-0.098^{***}	-0.100^{***}	-0.301^{***}	-0.143	-0.436^{***}	-0.340^{***}
	(0.016)	(0.024)	(0.024)	(0.023)	(0.020)	(0.110)	(0.113)	(0.151)	(0.124)
CBD 6	-0.291^{***}	-0.125^{***}	-0.094^{***}	-0.077^{***}	-0.090^{***}	-0.360^{***}	-0.135	-0.280^{**}	-0.142
	(0.018)	(0.019)	(0.017)	(0.016)	(0.016)	(0.100)	(0.089)	(0.130)	(0.116)
Inner Boundary 1–6			Yes	Yes	Yes		Yes		Yes
Outer Boundary 1–6			Yes	Yes	Yes		Yes		Yes
Kudamm 1–6				Yes	Yes		Yes		Yes
Block Characteristics					Yes		Yes		Yes
District Fixed Effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,260	6,260	6,260	6,260	6,260	5,978	5,978	2,844	2,844
R^2	0.26	0.51	0.63	0.65	0.71	0.19	0.43	0.12	0.33

Q: floor price. $EmpR$: employment by residence. $EmpW$: employment by workplace. CBD 1–6: six 500 m distance bands from the pre-war CBD. Inner/Outer Boundary and Kudamm: six 500 m distance bands from the respective boundary or Breitscheid Platz. Block characteristics: distances to schools, parks and water, block area, wartime destruction, land use, government buildings and urban regeneration policies. HAC standard errors (Conley, 1999) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

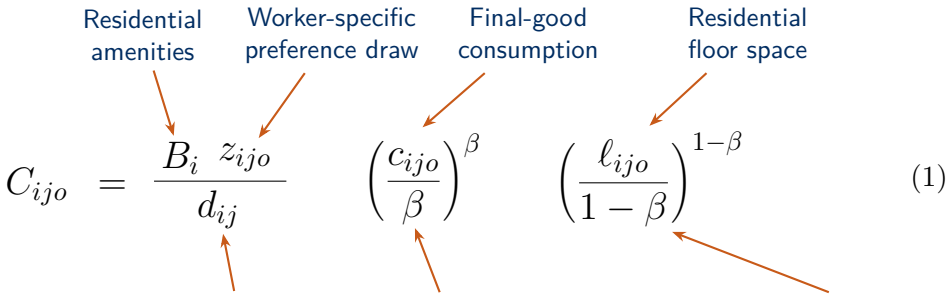
Table 2: Reunification, 1986–2006

	(1) $\Delta \ln Q$	(2) $\Delta \ln Q$	(3) $\Delta \ln Q$	(4) $\Delta \ln Q$	(5) $\Delta \ln Q$	(6) $\Delta \ln EmpR$	(7) $\Delta \ln EmpR$	(8) $\Delta \ln EmpW$	(9) $\Delta \ln EmpW$
CBD 1	0.398***	0.408***	0.368***	0.369***	0.281***	1.079***	1.025***	1.574***	1.249**
	(0.105)	(0.090)	(0.083)	(0.081)	(0.088)	(0.307)	(0.297)	(0.479)	(0.517)
CBD 2	0.290***	0.289***	0.257***	0.258***	0.191**	0.589*	0.538*	0.684**	0.457
	(0.111)	(0.096)	(0.090)	(0.088)	(0.087)	(0.315)	(0.299)	(0.326)	(0.334)
CBD 3	0.122***	0.120***	0.110***	0.115***	0.063**	0.340*	0.305*	0.326	0.158
	(0.037)	(0.033)	(0.032)	(0.032)	(0.028)	(0.180)	(0.158)	(0.216)	(0.239)
CBD 4	0.033***	0.031	0.030	0.034	0.017	0.110	0.034	0.336**	0.261
	(0.013)	(0.023)	(0.022)	(0.021)	(0.020)	(0.068)	(0.066)	(0.161)	(0.185)
CBD 5	0.025***	0.018	0.020	0.020	0.015	−0.012	−0.056	0.114	0.066
	(0.010)	(0.015)	(0.014)	(0.014)	(0.013)	(0.056)	(0.057)	(0.118)	(0.131)
CBD 6	0.019**	−0.000	−0.000	−0.003	0.005	0.060	0.053	0.049	0.110
	(0.009)	(0.012)	(0.012)	(0.012)	(0.011)	(0.039)	(0.041)	(0.095)	(0.098)
Inner Boundary 1–6			Yes	Yes	Yes		Yes		Yes
Outer Boundary 1–6			Yes	Yes	Yes		Yes		Yes
Kudamm 1–6				Yes	Yes		Yes		Yes
Block Characteristics					Yes		Yes		Yes
District Fixed Effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,050	7,050	7,050	7,050	7,050	6,718	6,718	5,602	5,602
R^2	0.08	0.32	0.34	0.35	0.43	0.04	0.07	0.03	0.06

Q: floor price. $EmpR$: employment by residence. $EmpW$: employment by workplace. CBD 1–6: six 500 m distance bands from the pre-war CBD. Inner/Outer Boundary and Kudamm: six 500 m distance bands from the respective boundary or Breitscheid Platz. Block characteristics: distances to schools, parks and water, block area, wartime destruction, land use, government buildings and urban regeneration policies. HAC standard errors (Conley, 1999) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model: Workers

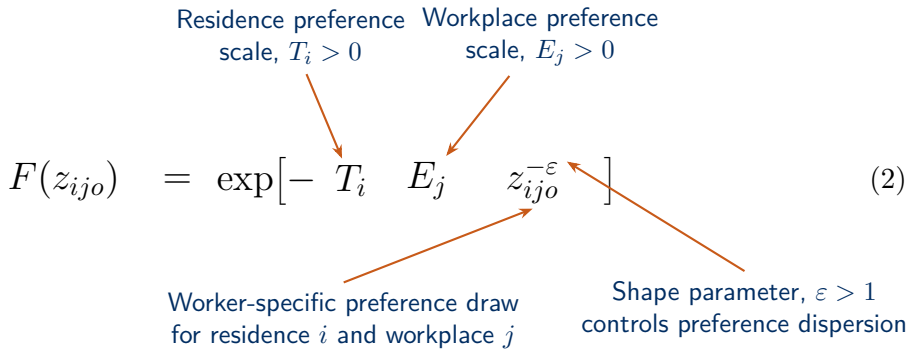
Consumption Index



i: residence block j: workplace block o: worker

Model: Workers

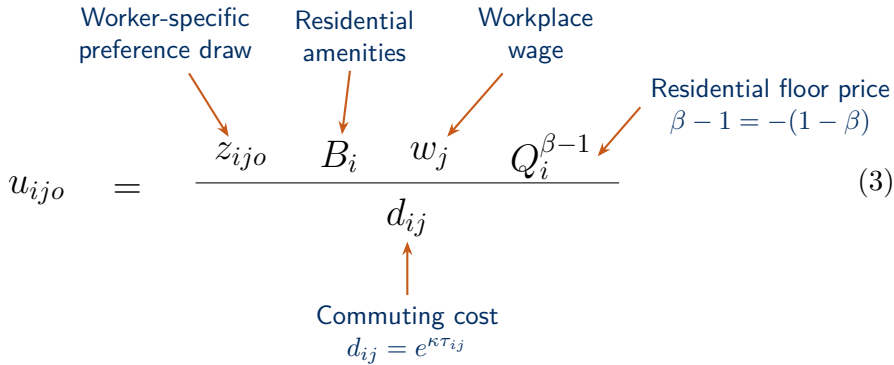
Idiosyncratic Preferences



Independent draws across residence–workplace pairs for each worker.

Model: Workers

Indirect Utility



i: residence j: workplace o: worker Final-good price normalised to 1

Model: Workers

Conditional Commuting Probability

Workplace preference scale

Wage relative to commuting cost

Fréchet shape parameter controls preference dispersion

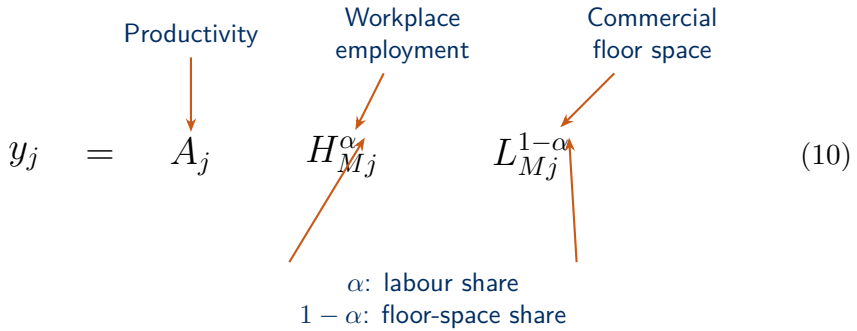
$$\pi_{ij|i} = \frac{E_j \quad (w_j/d_{ij})^{\varepsilon}}{\sum_{s=1}^{S} E_s (w_s/d_{is})^{\varepsilon}} \tag{6}$$

Sum across all S workplaces
s indexes alternative workplaces

Given residence i, Q_i, T_i and B_i cancel from the choice probability.

Model: Production

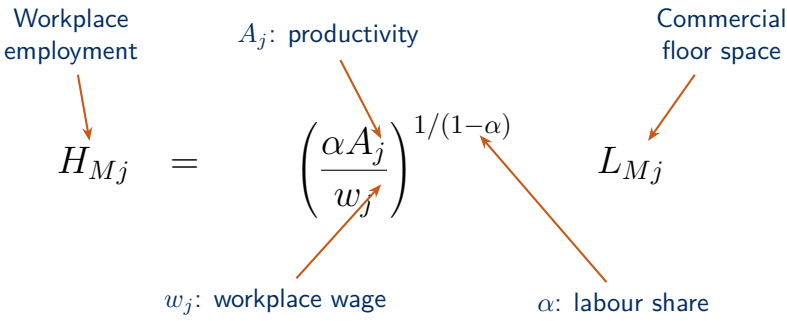
Production Function



Perfect competition and constant returns to scale.

Model: Production

Employment



Wages equate labour demand and commuting supply.

Model: Agglomeration

Production fundamentals

Travel-time decay

Workplace employment per unit of land

$$A_j = a_j \Upsilon_j^{\lambda}, \qquad \Upsilon_j = \sum_{s=1}^{S} e^{-\delta \tau_{js}} \frac{H_{Ms}}{K_s} \tag{20}$$

λ: production spillover strength

Residential fundamentals

Travel-time decay

Resident employment per unit of land

$$B_i = b_i \Omega_i^{\eta}, \qquad \Omega_i = \sum_{r=1}^{S} e^{-\rho \tau_{ir}} \frac{H_{Rr}}{K_r} \tag{21}$$

η: net residential spillover strength

A_j: productivity B_i: amenities δ, ρ: decay rates τ: travel time

Model: Solution

Given parameters and observed floor prices, workplace and residence employment, land area, and travel times:

1. **Recover adjusted wages $\widetilde{w}$.**
   Match commuting flows to observed workplace and residence employment.

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   Use adjusted wages, floor prices and firms' zero-profit condition.

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3. **Recover adjusted amenities $\widetilde{B}$.**
   Use adjusted wages, floor prices, residence shares and population mobility.

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4. **Separate fundamentals from spillovers.**
   Apply (20)–(21): $\widetilde{a}_j = \widetilde{A}_j / \Upsilon_j^{\lambda}$ and $\widetilde{b}_i = \widetilde{B}_i / \Omega_i^{\eta}$.

Model: Solution

Given parameters and observed floor prices, workplace and residence employment, land area, and travel times:

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5. **Recover adjusted development density $\widetilde{\phi}$.**
   Use land area and implied floor-space demands to clear the land market.

Model: Estimation

Estimation is by GMM using the Wall as exogenous variation.

I'm going to skip this.

Model: Results

Model: Results

Table 5, column 3: pooled division and reunification estimates

Parameter	Estimate (SE)	Meaning
$\nu = \kappa\varepsilon$	0.0987^{***} (0.0016)	How quickly commuting flows decline as travel time rises.
ε	6.6941^{***} (0.0934)	How similar workers' location preferences are. Higher values mean less dispersion.
λ	0.0710^{***} (0.0054)	A 1% rise in weighted workplace density raises productivity by 0.071%.
δ	0.3617^{***} (0.0782)	How quickly production spillovers fade with travel time.
η	0.1553^{***} (0.0083)	A 1% rise in weighted resident density raises amenities by 0.155%.
ρ	0.7595^{***} (0.1741)	How quickly residential spillovers fade with travel time.

HAC standard errors in parentheses. *** significant at 1%. Travel time is measured in minutes.

Model: Results

Table 6: externalities and commuting costs

Travel time	Production externalities $e^{-\delta\tau}$	Residential externalities $e^{-\rho\tau}$	Utility after commuting $e^{-\kappa\tau}$
0 minutes	1.000	1.000	1.000
1 minute	0.696	0.468	0.985
2 minutes	0.485	0.219	0.971
3 minutes	0.338	0.102	0.957
5 minutes	0.164	0.022	0.929
7 minutes	0.079	0.005	0.902
10 minutes	0.027	0.001	0.863
15 minutes	0.004	0.000	0.802
20 minutes	0.001	0.000	0.745
30 minutes	0.000	0.000	0.642

Spillovers are highly local. Commuting utility declines much more slowly.

Model: Results

Table 7: simulated versus observed

	Division		Reunification	
Distance band	Observed	Simulated	Observed	Simulated
CBD 1 (nearest)	−0.800	−0.836	0.398	0.363
CBD 3	−0.543	−0.455	0.122	0.163
CBD 6	−0.291	−0.349	0.019	0.100

Effects on log floor prices in selected 500 m distance bands.
Simulations use the baseline model with both spillovers.

Extensions of the Framework

- **Monte, Redding and Rossi-Hansberg (2018), AER**
  Commuting, Migration, and Local Employment Elasticities
  Brings in more complex labour market structure and studies local shocks.
- **Heblich, Redding and Sturm (2020), QJE**
  The Making of the Modern Metropolis: Evidence from London
  Uses a QSM to study the rollout of railways in Victorian London
- **Couture, Gaubert, Handbury and Hurst (2024), REStud**
  Income Growth and the Distributional Effects of Urban Spatial Sorting
  Adds income heterogeneity and endogenous amenities.
- **Tsivanidis (2026), AER**
  Evaluating the Impact of Urban Transit Infrastructure: Evidence from Bogotá's TransMilenio
  Adds worker groups and transit modes to evaluate welfare and distribution.
- **Robert-Nicoud, Combes, Duranton & Gobillon, Working Paper out a week ago**
  A Unified Urban Model with Non-Homothetic Housing Demand
  Brings in more realistic housing demand which explains varying income shares spent on housing and the distribution of city sizes.